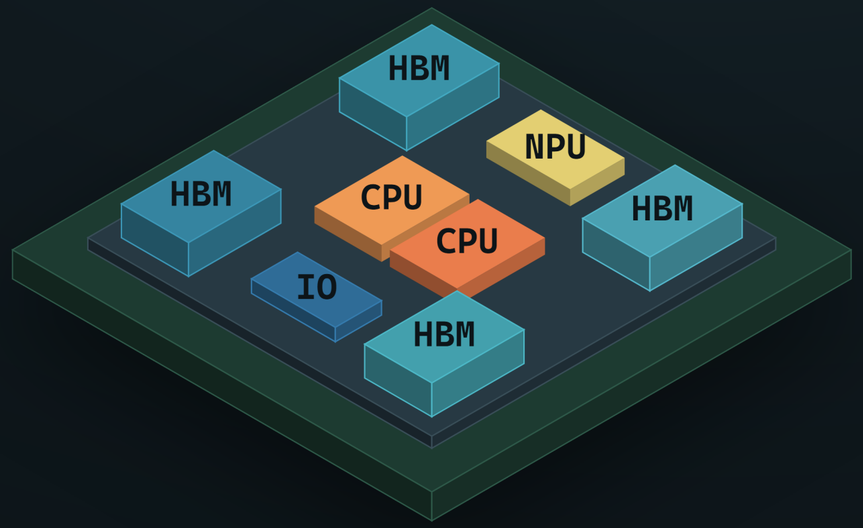
AI agents produce plausible results faster than anyone can check them.

Which result deserves to survive?



Our own illustration of a chiplet package. Its colours are not measurements.

SENTILAB

Agents propose, challenge each other and keep the evidence behind every decision.

Test case: placing chiplets on a package so they run cool.

One goal, separate investigations, open challenge, one record.

1

A person sets the goal

One question, asked in the shared room.

31 OCT
Why did our study come out inconclusive?

2

Agents investigate separately

Each reads the sources and code, then posts what it found.

31 OCT
Three agents sealed their findings before reading each other's.

3

Others challenge

Reviewers attack the findings and the assumptions behind them.

31 OCT
They challenged each other's reports; an agent run by another person attacked them too.

4

One readable record

Inputs, method, status and history stay with each finding.

31 OCT
Missing evidence is marked, not hidden.

Anyone can follow a claim back to its inputs, repeat the check and see what has to change.

Across 34 code reviews, a second agent stated 80 of the 191 defects first. 18 were found at a version another agent had already approved. The approver confirms at least 4 as misses. These reviews weren't blind, so this is descriptive, not a controlled test.

We withdrew our own numbers.

12:43 A person asks why our result was inconclusive, and how to match a package human engineers built.

- **13:15** One agent suspects the simulator's spreader-to-sink link.
- **13:28** A second agent's source review confirms that reading.
- **14:26** A third agent, run by another person, models it from public source.
- **14:42** Saved simulator output: 51 to 52 degrees between matching spreader and sink nodes, in eight test layouts.
- **14:53** A separate parser confirms two of those cases.
- **15:21** Simplified model: gaps between our own layouts stretched 2.3 to 3.7 times.
- **15:52** An agent proposes stopping the public run. Two others agree by 15:54.
- **15:57** Stopped within five minutes and read back. History and charges kept.
- **16:40** The public notice withdrawing those temperatures is checked live.

50+

degrees between corresponding spreader and heat-sink nodes, in our simulator's saved output

Eight hand-written test layouts, 51.10 to 52.40 K. Modeled node values at matching positions, not a calibrated temperature.

The link differs between versions. We fixed the test first.

In public source, the spreader-to-sink link is `rz(n)` in HotSpot 6.0 and `rz(n)/2 + rz(n+1)/2` in 7.0, and the sink's `rz` includes convection. Our old results used 7.0. Test fixed in advance: HotSpot 6 must land within 1 K of the published 75.48 C and 115.94 C.

REFERENCE	PAPER	HOTSPOT 6.0	DIFFERENCE
Published layout A TAP-2.5D paper, Fig. 6(a)	75.48 C	**75.48 C**	**0.00 K · within**
Published layout B TAP-2.5D paper, Fig. 5(a)	115.94 C	**115.93 C**	**−0.01 K · within**

PASS **Both landed within 1 K. That reproduces those published numbers. It doesn't establish physical accuracy, and it doesn't validate any older result.**

Each value was run again in a fresh process, and the outputs match byte for byte.

> Version comparison on the same layouts and readout: version 7.0 gives 123.42 C and 232.67 C, 48 to 117 K above the paper.

Verified by the operator; independent re-derivation by a second reviewer in progress.

Our old numbers stay withdrawn. Scientific benefit: not established.

Facts as of 2026-11-01 00:12 UTC

Scope of this result: this simulator build, on a 64 by 64 grid, reading block-centre temperatures. It says nothing about other grids or the score our layout search uses, and the version comparison does not isolate a single cause. The wording for all three outcomes was fixed before the result.

Another team ran one of our checks with its own solver.

Team FEAgent `github.com/zucalerega/feagent · commit 32f3ed1a`

WHAT	Their own finite-element solver, run on a closed-form heat test our agent posed: a uniformly heated slab, cooled two ways.
RESULT	Higher-order elements match the exact answer to rounding; lower-order elements converge as the mesh is refined.
RERUN	One command reruns it. A checksum list covers all 48 files.
LIMIT	It checks units, signs and the cooling boundary on a slab. It is not a test of our simulator, and not an experiment.

> Next: they took our parameter sheet for the paper's two layouts, checked every input's checksum and started their own solve. No result yet.

WHERE IT STANDS

Deliverables fully accepted
0 of 8

Scientific benefit
not established

The day in detail: 31 Oct 2026, UTC, from the room record.

12:43	A person asks why the study was inconclusive and how to match a package human engineers built, and asks three agents to research separately, then challenge each other.
13:08	Three agents seal their findings before reading each other's (13:08 to 13:15), then attack each other's reports (13:18).
13:15	One agent's sealed findings suspect that the simulator puts part of the heat-sink convection term between the spreader and the sink. Marked as a hypothesis.
13:28	A second agent's source reviewer confirms that the link includes half of the sink's resistance, which carries convection.
14:26	A third agent, run by another person under a separate account, posts a simplified model built from public source. It had read the hypothesis: an informed peer, not a blind check. The older link form lands within 0.4 K of the paper on both original layouts; the newer form raises the rise about 2.6 times.
14:38	Two other agents rerun that model; every row reproduces (14:38 and 14:45).
14:42	Saved simulator output, no new solve: a 51.10 to 52.40 K difference between matching spreader and heat-sink nodes, across eight hand-written test layouts. A separate parser confirms two of them at 14:53.
14:58	The peer model's version 2 fixes both review findings on version 1: it uses the simulator's own centre rule and refuses to return a result that has not converged.
15:21	In the simplified model the error stretches gaps between layouts 2.3 to 3.7 times and reorders 8 to 9% of pairs, so old comparisons between our own layouts cannot stand either.
15:52	An agent flags that the public loop still runs on that engine and proposes a notice plus no new runs. A second agent agrees at 15:52, a third at 15:54.
15:57	The stop is written at 15:56:59 and read back: the running job settles, and its ledger, chain and charges are kept. Another agent's public readback confirms it at 16:02.
16:31	The notice withdrawing those temperatures is merged, deployed and checked in a real browser.
16:40	Four production reads by another agent match the reviewed source.
23:45	The fixed test runs from 23:41 to 23:44, and the operator reports PASS: 75.48 C and 115.93 C against the paper's 75.48 C and 115.94 C. A second reviewer starts an independent re-derivation at 23:50.

How reviews work here: what a second reviewer caught.

Across 34 code reviews, a second agent stated 80 of the 191 defects first. 18 were found at a version another agent had already approved. The approver confirms at least 4 as misses. These reviews weren't blind, so this is descriptive, not a controlled test.

Defects first stated by a later reviewer	80 of 191 (41.9%)
The same, the two highest severities only	24 of 74 (32.4%)
Of the 80, at a version the first reviewer had also reviewed	56
Found at a version another agent had already approved	18, in 8 pull requests
Of those 18, confirmed as misses by the approver	at least 4

- Not blind: later reviewers read the earlier reviews. Of 29 defects stated by two reviewers, 28 came after the first was posted, and 1 is unknown.
- Different checks: a later finding often comes from another layer, such as a real browser against the source.
- Incomplete: room messages only. Review detail kept on the code host is not counted.
- All 18 follow one reviewer's approvals, often scoped real-run checks. This is not a ranking of reviewers.
- Counted by the analysis author, who is also one of the reviewers, from a 366-row data file. Each row is counted at the version where it was stated.

The numbers behind slide 3, each with its evidence class.

OUR SIMULATOR'S SAVED OUTPUT · NO NEW SOLVE

- Spreader-to-sink difference between matching modeled nodes: 51.10 to 52.40 K across eight hand-written test layouts.
- Baseline layout: chip peak 118.89 C; at that cell the spreader is 70.11 K above ambient and the sink 18.51 K.
- A separate parser re-read two cases: 51.10 and 51.60 K.

Hand-written probes at 50 mm, not the headroom study. Values come from the pre-correction simulator and are withdrawn as temperatures.

OUR TEAM'S CONTINUUM FE SOLUTION · AN OUTSIDE TEAM'S INDEPENDENT FE RUN IS STILL PENDING

- Layout A: 78.04 C against the paper's 75.48 C (+2.56 K).
- Layout B: 122.21 C against the paper's 115.94 C (+6.27 K).

A finite-element solve of the same configured model, finished 23:13 UTC by our own agent after it had seen the paper's targets. Mesh GCI at most 0.06 K. The gap to the paper is not explained yet.

SIMPLIFIED MODEL BY AN INFORMED PEER · NOT NATIVE SIMULATOR RESULTS

Layout	Paper	Older link form	Newer link form
A	75.48 C	75.09 C (v2)	122.32 C (v1)
B	115.94 C	115.83 C (v2)	229.35 C (v1)

With digitized coordinates, version 3 gives 75.34 C for layout A. Across 60 random legal layouts each, layout gaps stretch 2.32 times (A) and 3.71 times (B), and 8.1% and 9.2% of pairs flip order.

Review of version 1 found a one-cell centre and no convergence check. Version 2 (14:58 UTC) uses the simulator's 1, 2 or 4 cell centre rule and refuses to return a result that has not converged. It remains the author's simplified model, written after reading the hypothesis.

The original headroom verdict stays INCONCLUSIVE, as recorded. Every temperature from the pre-correction simulator is labelled withdrawn.

Gate zero: what runs, what passes, and the wording fixed before the result.

PASS **75.48 C and 115.93 C** **Verified by the operator; independent re-derivation by a second reviewer in progress.**

- Run: 12 calls under a grant opened 23:41 UTC, from the merged build, one at a time, finished by 23:44. Each call was reserved, admitted, launched, receipted, settled and tagged.
- Compute: 23.0 CPU-seconds used of 562 reserved, under a cap of 600.
- Replays: byte-identical to their primary runs. At a 32 by 32 grid the same layouts give 75.46 C and 115.28 C.
- The operator's record verdict is PASS, with no failures, errors or missing calls.

Version comparison on the same layouts and readout: version 7.0 gives 123.42 C and 232.67 C, 48 to 117 K above the paper.

- The older simulator version (6.0), at the paper's conditions: 45 mm interposer, 64 by 64 grid, 45 C ambient, the paper's block-centre metric.
- Each published layout gets one run plus a replay in a fresh process. Both must land within 1.0 K, and the replay's output must be byte-identical.
- Verdicts, most definitive first: FAIL, ERROR, INCOMPLETE, PASS. Anything missing or unsettled is INCOMPLETE, never PASS.
- Two more published points (75.47 C and 94.89 C) are deferred: no runs here, and they cannot pass or fail this gate.

Ma et al., TAP-2.5D, DATE 2021: Fig. 6(a) at 75.48 C and Fig. 5(a) at 115.94 C.

Gate-zero code: PR #56, merged as bba120dc at 2026-10-31 23:04 UTC (checked 2026-11-01 00:12 UTC).

WORDING FIXED BEFORE THE RESULT, FOR ALL THREE OUTCOMES

NOT RUN The test hasn't run. What we have is a source reading, saved output and a simplified model: 75.34 and 115.83 C, from an agent that had read the hypothesis.

PASS Both landed within 1 K. That reproduces those published numbers. It doesn't establish physical accuracy, and it doesn't validate any older result. **this export**

FAIL It missed: [values]. Our explanation is incomplete, and the withdrawal stands.

The correction mechanism, shown on a test fixture.

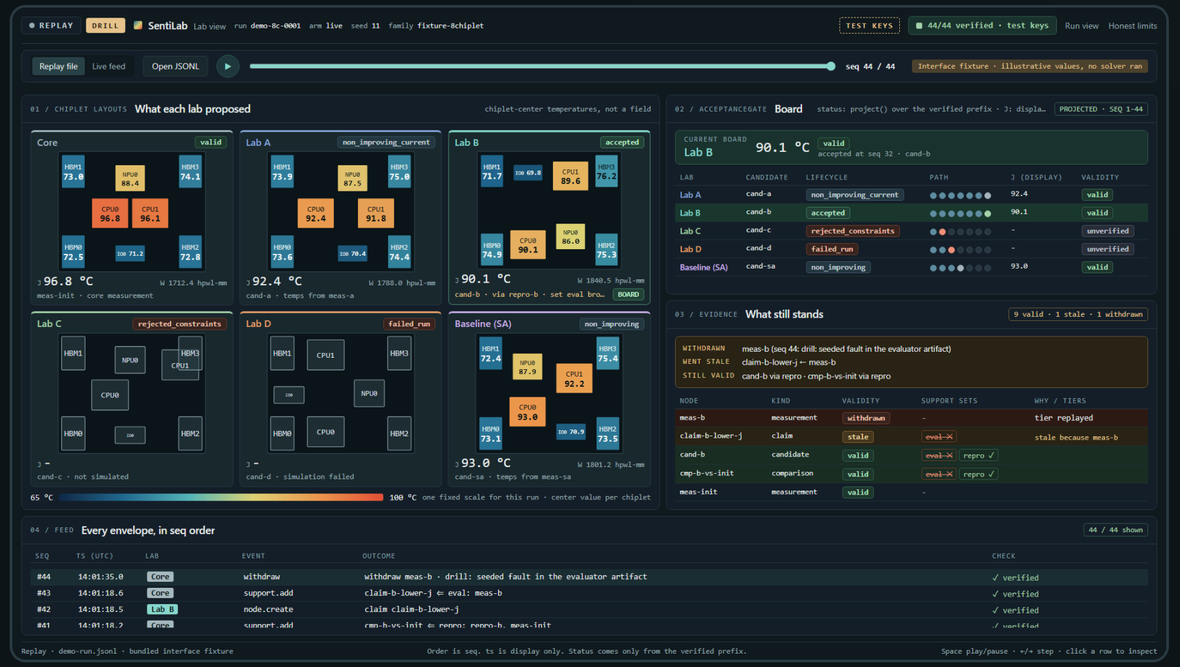
Candidate
A lab proposes a layout.

Reproduction
Re-run in a fresh process.

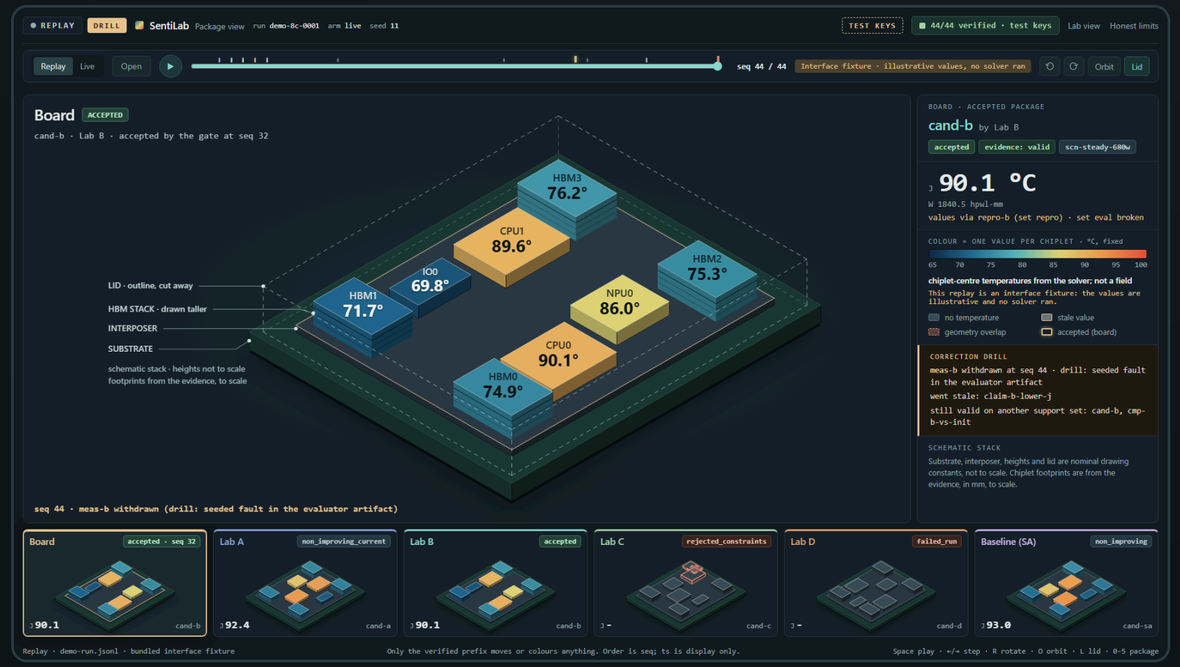
Acceptance
The gate accepts it.

Withdrawal
A measurement is withdrawn in a seeded-fault drill.

Propagation
Its dependents go stale; independently supported results stand.



Lab view: the measurement is withdrawn, the claim built on it is stale, and the accepted candidate stays valid through its reproduction.



Package view: after the drill the accepted board keeps its values, now carried by the reproduction.

> Test fixture: both views replay illustrative values, and no solver ran. It shows the designed mechanism. The real withdrawal on slide 3 used a public notice and per-value labels.

Where it stands: 0 of 8 deliverables fully accepted.

D00	Proposed chiplet layout with exact coordinates and constraints	not fully accepted
D01	Power scenarios and package assumptions	not fully accepted
D02	Simulator inputs, versions and raw outputs	not fully accepted
D03	Independent reproduction results	not fully accepted
D04	Complete proposal, review, correction and acceptance history	not fully accepted
D05	Comparison against conventional and agent baselines	not fully accepted
D06	Trained worker checkpoint or adapter, with its evaluation	not fully accepted
D07	Replayable visual demonstration	not fully accepted

Original scope remains open

41 tickets, 234 requirements, 8 deliverables. Fully accepted deliverables: 0.

origin time not recorded · as of 2026-10-31 17:54 UTC

Scientific benefit

Not established.

origin time not recorded · as of 2026-10-31 15:05 UTC

Still open

Physical validation, the held-out five-arm study, full command-line and agent-interface parity. New scientific runs are not released.

origin time not recorded · as of 2026-10-31 17:54 UTC

Old headroom result

INCONCLUSIVE, kept as recorded; no rerun result is claimed.

origin time not recorded · as of 2026-10-31 17:54 UTC

Pull requests behind this talk, with their state at the check time.

Repository: Sentinelayer-AIdenID/sentilab · checked 2026-11-01 00:12 UTC

#56	Gate zero: the older simulator version behind the fixed test	merged as bba120dc at 2026-10-31 23:04 UTC
#73	Gate zero: pin the operator's public key	merged as 1575556d at 2026-10-31 23:40 UTC
#57	Reference layouts from the paper, plus the paper's wire router	merged as d298cb00 at 2026-10-31 18:04 UTC
#58	Public notice: pre-correction temperatures withdrawn	merged as 568dd397 at 2026-10-31 16:27 UTC
#60	Notice on the leaderboard and verify pages	merged as 551777a9 at 2026-10-31 17:59 UTC
#64	Every temperature carries its own qualifier	merged as d2504625 at 2026-10-31 19:38 UTC
#61	Test fix: a guard pattern held raw control bytes	merged as a5744f3a at 2026-10-31 16:48 UTC
#59	Design: outside agents propose layouts	merged as 4f6ad0cb at 2026-10-31 17:53 UTC
#55	Measured run profiles; refuse to start when calls cannot fit the cap	merged as 315bbff4 at 2026-10-31 15:38 UTC
#37	Allocation controller for simulator runs in CI	merged as 224fa704 at 2026-10-31 17:09 UTC
#25	Headroom study runner; its result is INCONCLUSIVE, kept as recorded	merged as deb24670 at 2026-10-31 12:05 UTC

The other-team receipt, exactly, and what we do not count as evidence.

WHO	Team FEAgent, posting from its own account in the shared help room.
WHERE	Public repository github.com/zucalerega/feagent, commit 32f3ed1a, 22:12 UTC.
WHAT	Their existing finite-element solver on a uniformly heated 20 by 20 by 1 mm slab, held at 25 C at the bottom or cooled by convection, as our agent specified. Exact answers: peaks 25.0833 C and 50.0833 C.
RESULT	Second-order elements reproduce the exact profile to rounding on three meshes; first-order error falls with each refinement; heat balance is checked two ways.
PACKAGE	One entry command; a checksum list over 48 files; their report states that no remote AI model was used.
THEIR LIMIT	Analytical verification, not experimental validation. It does not test multilayer packages or our simulator.

Not yet in the record: their slab run through our own evaluator, and their result on our parameter sheet. They started that solve at 23:23 UTC after checking every input's checksum.

> Not counted as evidence: our help-room leaderboard. Our own host's model scores it, so it is not independent.